

Soumik Sarker

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Research Interests

- Human-centered Security | AI Governance (Model Monitoring & Bias Reduction)
- Privacy Preserving ML in Health | Secure Software Engineering & vulnerability mitigation

Education

Shahjalal University of Science and Technology (SUST)

B.Sc in Computer Science and Engineering

CGPA - 3.60/4.0

Sylhet

2017 - 2021

Publications

Improving joint attention in children with autism: a VR-AR enabled game approach

- An AR/VR game strategy to solve real-life **social interaction problems** for kids, struggling with joint attention.
- Architecture is designed efficiently resolving the eye strain and other health risks from the direct involvement of the devices in the strategy
- [Publication Link](#)

A Survey on Blockchain & Cloud Integration

- A thorough review of **blockchain-integrated services** in cloud platforms, specifically focusing on IaaS, PaaS, and SaaS models.
- A categorical representation of these integrations into four sections: **Security, Blockchain, Federation, and Management** -as a Service.
- Evaluation of each work using standardized metrics, feature comparisons, and constraint analysis.
- [IEEE Publication Link](#) | [Presentation Link](#)

Word Completion and Sequence Prediction in Bangla Language Using Trie and A Hybrid Approach of Sequential LSTM and N-gram

- A hybrid linguistic model integrating **N-grams** and **bidirectional LSTM** for enhanced auto-completion and sequence generation.
- Architecture is designed to consider the unique characteristics of the **Bengali Language** while mitigating the limitations of existing architectures.
- Curation of a distinctive **dataset**, characterized by its rich vocabulary, drawn from **diverse literary sources** and Bangla news portals.
- Demonstration of superior performance, especially in scenarios requiring **preference-specific word completion** or **sentence suggestion**.
- [IEEE Publication Link](#) | [Presentation Link](#)

Deep Learning Approach Combining Lightweight CNN Architecture with Transfer Learning: An Automatic Approach for the Detection and Recognition of Bangladeshi Banknotes

- A performance enhancement experiment using **lightweight CNN** models for automatic banknote detection and recognition.
- Experimented with different transfer learning and lightweight models like **MobileNet** and **NASNetMobile**.
- Combination of **data augmentation techniques**, **hyperparameter tuning**, and **optimization methods** for resolving limited dataset issues.
- Performance evaluation and demonstration of superior accuracy rates, outperforming the heavier **ResNet152v2** model.
- [IEEE Publication Link](#)

Work Experience

Therap (BD) Ltd.

Software Engineer II

Dhaka

Jul 2021 - Present

Therap is a US-based software company supporting health-specific managerial solutions for individuals, service providers, and states.

- Contributed to several core systems maintaining engineering strategies like **Stateless Auth Bridging**, **Declarative Authorization**, **Zero-Trust Security Framework**, **Rule-based Guardrails**
- Involved in several **R&D projects** i.e. **Therap Report Lib.** (data export as pivot or spreadsheet, visualization tool), **Therap DOCx** (custom document processor), **Authentication Service**, Integration with IVR **PlumFuse**
- Took an active role in several significant releases of our system like **JDK**, **Spring**, **jQuery** and **Bootstrap** version upgrades.
- Handled several web security, **vulnerability & edge hardening** scopes i.e. **CSRF Security**, URL authentication with **HMAC Signature**, **Content Security Policy**
- Learned about security concerns and practices while developing external **JARs** or integrating with third-party libraries.

BRAC HNPP APPLICATION DEVELOPMENT

Developer and Content Designer

Brac Center, Dhaka

Oct 2019 - Nov 2020

A contractual work with **Brac HNPP**, worth **4.3k USD**, for developing trainee (health workers in the training phase) management alongside a learning module.

- Developed a **learning and management tool** for health workers of the mentioned organization.
- Handled successful application maintenance (**comprising 25% of the current user base**) for a considerable time.
- Worked in a cross-functional environment and communicated with client requirements through several field visits.
- Currently on-live with more than **3k plus** users.

Virtual Reality (VR) based E-learning application for children with ASD

SUST Research Center, SUST, Sylhet

Research Assistant and Developer

Jul 2019 - Jun 2020

University research center funded our work on the implication of VR-based interactive module feasibility and acceptability for special children.

- Proposed an interactive **VR-based module** for people (especially children) with **Autism Spectrum Disease**.
- Navigated existing VR-related approaches and developed a prototype based on our proposition .
- Designed and developed interactive app integrated with **Augmented Reality** and **Virtual Reality**.
- **Supervisor:** [Md Mahadi Hasan Nahid](#) | [Project Paper](#)

Thesis & Notable Projects

Security-as-a-Service (SaaS): A Three-way Blockchain Based Security Service Delivery Model for Cloud

[Thesis Report](#)

I proposed a **three-way blockchain-enabled security service** (Access control, Secure Searchable Encryption, and Data provenance) on a cloud platform.

- Implemented blockchain-enabled **Secure Searchable Encryption (SSE)** on cloud platform.
- Data owners on cloud can have full secured control and shared policies over the files.
- Data owners can customize **access policies** through the use of **smart contracts**.
- By using **STRIDE threat modeling**, I verify the blockchain network for proper judgment and efficiency calculation.
- Study existing blockchain implementation on cloud in all sorts of scopes.
- **Platform:** Hyperledger Fabric, AWS Cloud
- **Supervisor:** [Dr. Md Sadek Ferdous](#) | [Thesis Presentation Link](#)

SHIKHA (Torch - A light to freedom)

[Github Link](#)

A story-based short-length role-playing game (RPG) based game.

- Players have to pass different obstacles and solve various sorts of riddles.
- I developed all the game environments, objects, and sprites myself.
- **Tools:** Unity, C#, Adobe Illustrator, Coral Essentials

SUST ONLINE EXAM

[Github Link](#)

A website to conduct exams and monitor student progress online.

- Manage new courses, course material, exams, exam questions and answers, and student performance from the teacher's side.
- Join an assigned exam from a particular teacher and get the result from the student's side.
- **Tools:** PHP, Laravel, MySQL

Skills

	Languages	Java (8, 17, 21), JavaScript, TypeScript, Python, SQL
Backend Frameworks & Libraries	Backend Frameworks & Libraries	Spring Core-Security, Hibernate ORM, JPA, MyBatis, Enterprise Java Beans, LMAX Disruptor, Log4j2
	Frontend Technologies	React 18, Custom Web Components (Shadow DOM), Redux/RTK, Vite, Webpack, Bootstrap 5, jQuery
	Database	MySQL, MongoDB, Oracle SQL
	Distributed Systems & Caching	Hazelcast (IMDG), Oracle Coherence, WebLogic JMS (Uniform Distributed Destinations)
	Web & Application Servers	Apache Tomcat (9, 11), Oracle WebLogic (14c), Nginx
	Security & Cryptography	JWT, SAML 2.0, Bouncy Castle Cryptography, PGP, HMAC, OWASP Protocols (CSRF, XSS, CSP mitigations)

Achievements & Roles

2019	WINNER (Team Nrick) BRACATHON 3.0
	Designed and developed a prototype of a digitalized learning approach for the health workers of BRAC .
2019	WINNER (Team Nrick) SUST TechFest
	Presented a VR-based learning module for Autism of several age stages. Through this application, we focused on Joint Attention difficulty, a common issue for people with Autism.
2018	1st RUNNERS-UP (Team Nrick) IUT 10th ICT Fest
	Developed a prototypical app for patients seeking the right doctors, AI-based guidance , and emergency help .
2019-2020	Secretary General SUST ACM Student Chapter
	Organized several coding events , workshops , and especially virtual sessions during the COVID lockdown.
2019	Head of Design ICBSLP 2019
	Supervised publication team of the International Conference on Bangla Speech and Language Processing.

Open-Source Project Contributions

- I have been actively contributing (**bug detections**, **feature fixes**, and **code reviews**) to several key open-source projects (i.e., [Spring Security](#), [Flower Federated Framework](#)) on [GitHub](#). Details can be found [here](#).